Computer Vision Metrics

1 Introduction

Here we present a number of tasks in computer vision and detail the metrics applicable to them. The tasks are organized into categories. Within each metric-applied-to-task there are four points that should be covered, as shown in the example in Section 1.1.1. Comments can be made using the `\textcomment{}` command like so: **Comment: Comments can be used for ideas or references to papers.**

1.1 Example CV Task

Basic task description.

1.1.1 Metric

Details of the metric as applied to the given CV task. Points that should be covered:

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2 Image Analysis Tasks

2.1 Object localization

Locating and demarcating an instance of an object within an image, using for instance a bounding box.

2.1.1 Intersection-over-Union

a. *Metric definition and/or formula:*

   See formula in A.3

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.1.2 Box Intersection-over-Union

a. *Metric definition and/or formula:*

   Box-IoU is IoU when the predicted object and ground truth reference objects are delimited by bounding boxes. The metric is calculated using the formula in A.3.

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.1.3 Mask Intersection-over-Union

a. *Metric definition and/or formula:*

   Mask-IoU is IoU when the predicted object and ground truth reference objects are delimited by pixel masks. The metric is calculated using the formula in A.3.

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.1.4 Boundary Intersection-over-Union

a. *Metric definition and/or formula:* Instead of using the full areas of the predicted object and ground truth reference objects Boundary IoU uses only the areas within a certain distance d of the borders of each object.

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:* Parameter d, the distance from the object boundary that should be considered for the metric calculation.

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.2 Object classification

Assigning a label to an object in an image.

2.2.1 Accuracy

a. *Metric definition and/or formula:*

See formula in A.1.1

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.2.2 Precision, recall, specificity, Negative Predictive Value

a. *Metric definition and/or formula:*

See formulas in A.1.2 for precision, in A.1.3 for recall, in A.1.4 for specificity, in A.1.7 for Negative Predictive Value

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.2.3 F_1-score, F_β-score

a. *Metric definition and/or formula:*

See formula in A.1.8 for F1, in A.1.9 for F_β

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

For F_β: β (see A.1.9)

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.2.4 Balanced accuracy, Matthews Correlation Coefficient, Positive Likelihood Ratio, Negative Likelihood Ratio, Youden's Index

a. *Metric definition and/or formula:*

See formula in A.1.10 for balanced accuracy, in A.1.11 for Matthews Correlation Coefficient, in A.1.12 for Positive Likelihood Ratio, in A.1.13 for Negative Likelihood Ratio, in A.1.14 for Youden's Index

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.3 Object classification with confidence

Assigning a label and probability to an object in an image.

2.3.1 Expected Cost

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.3.2 AUROC

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.3.3 Sensitivity@Specificity

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.3.4 Expected Calibration Error

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.3.5 Maximum Calibration Error

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.3.6 Brier Score

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.4 Object detection

Detecting one or multiple instances of a given object class in an image, outputting for each instance its bounding box and its label. Requires both localizing the instance within the image, and determining the label of the instance.

2.4.1 Precision, Recall

a. *Metric definition and/or formula:*

   See formula in A.1.2 for precision, in A.1.3 for recall

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

   - Assignment strategy (for pairing each predicted detection with a ground truth object, if there are several), such as:
     - Greedy by Score
     - Hungarian Matching
   - Criterion to determine if the predicted detection / ground truth object matched pair constitutes a successful match (hit, TP) or not (miss, FP), such as:
     - Center-cover criterion: 1 (hit) if the center of the ground truth reference object is within the bounding box or mask of the predicted detection; 0 (miss) otherwise. *Caveat: Calculating the center point for the ground truth object is simple for a bounding box but more complicated for a mask.*
     - Center-hit criterion: 1 (hit) if the center of the predicted object is within the bounding box or mask of the ground truth reference object; 0 (miss) otherwise. *Caveat: Calculating the center point for the predicted object is simple for a bounding box but more complicated for a mask.*
     - Distance-based hit criterion: 1 (hit) if the distance d between the bounding box or mask centers of the ground truth reference object and the predicted object are within distance threshold τ; 0 (miss) otherwise. *Caveat: Calculating the center points for the ground truth object and the predicted object is simple for a bounding box but more complicated for a mask.*
     - Intersection-over-Union (IoU), as defined in A.3, with a threshold above which the pair is considered a successful match
     - Box IoU (see 2.1), with a threshold above which the pair is considered a successful match
     - Mask IoU (see 2.1), with a threshold above which the pair is considered a successful match
     - Boundary IoU (see 2.1), with a threshold above which the pair is considered a successful match

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.4.2 F_1-score, F_β-score

a. *Metric definition and/or formula:*

   See formula in A.1.8 for F1, in A.1.9 for F_β

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

   - For F_β: β (see A.1.9)
   - Same as for precision and recall in 2.4.1

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.5 Object detection with confidence

Detecting one or multiple instances of a given object class in an image, outputting for each instance its bounding box, its label and a confidence score.

2.5.1 Expected Cost

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.5.2 Expected Calibration Error

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.5.3 Maximum Calibration Error

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.5.4 Brier Score

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.5.5 Mean Average Precision

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.5.6 Free-Response Receiver Operating Characteristic (FROC), FROC Score

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.6 Semantic segmentation

Basic task description.

2.6.1 Dice Similarity Coefficient

Also known as: Sørensen-Dice Index; Sørensen Index; Dice's Coefficient; F_1-score when applied to Boolean data.

a. *Metric definition and/or formula:*

$$DSC = \frac{2\,|X \cap Y|}{|X| + |Y|} \tag{1}$$

where X and Y are sets with cardinalities (i.e. number of elements) $|X|$ and $|Y|$ respectively.

In the context of semantic segmentation the sets X and Y will respectively be the predicted and ground truth pixels for a single class.

When using multiple classes the DSC should be calculated for each class and then averaged together [1]. If instead the classes are combined together before the calculation the result will be biased towards the more prevalent class [3].

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.6.2 Jaccard Index

Also known as: Intersection over Union (IoU)

a. *Metric definition and/or formula:*

See formula in A.3.

An alternative formulation following [2] is as follows: let n_{ij} be the number of pixels of class i that are labeled as class j (thus making n_{ii} the number of correctly labeled pixels of class i). Then let $t_i = \sum_j n_{ij}$ be the total number of pixels of class i. The Jaccard Index/IoU for one class is thus

$$\mathrm{IoU}_i = \frac{n_{ii}}{t_i + \sum_j n_{ji} - n_{ii}} \tag{2}$$

Across multiple classes the **mean intersection over union** is defined as

$$\text{mean IoU} = \frac{1}{n_{\mathrm{cl}}} \sum_i \frac{n_{ii}}{t_i + \sum_j n_{ji} - n_{ii}} \tag{3}$$

where n_{cl} is the number of classes. Open questions:

- Is this calculated per image, and then averaged? Or are all the pixels from all the images in the set combined and the calculation done over all of them simultaneously?
- Is n_{cl} the number of classes present in the ground truth labels? Or the combination of the ground truth labels and the predicted labels (i.e. what if the SiS system has a wider variety of labels than the dataset)?
- How are multiple segmentation labels handled?

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.6.3 Pixel Accuracy

a. *Metric definition and/or formula:*

$$\text{Pixel Accuracy} = \frac{\text{Number of correctly labeled pixels}}{\text{Total number of pixels}} \tag{4}$$

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.6.4 Haussdorff Distance aka Maximum Symmetric Surface Distance

a. *Metric definition and/or formula:* The most general version of the Hausdorff Distance is defined over two non-empty subsets of a metric space. It can thus be defined in N-dimensional Euclidian space. Here we assume a two dimensional Euclidean space, and so use the term pixel to denote the subelement of a the 2D data array that is the image, but the formulation would work equally well for voxels of 3D data arrays, or hypervoxels in N-dimensional data arrays where $N > 3$.

   The distance between a pixel a and a set of pixels B is first defined as

$$d(a, B) = \min_{b \in B} d(a, b) \tag{5}$$

   where $d(a, b)$ is the Euclidean distance between two pixels incorporating the real spatial resolution of the image [4]. *Question: this would thus require depth information; and the calculation would be simple for relatively flat images e.g. dermatology images, but much more difficult for those with wider depth distributions.*

$$d_{\mathrm{H}}(X, Y) = \max \left\{ \max_{x \in \partial X} d(x, \partial Y), \; \max_{y \in \partial Y} d(X, \partial y) \right\} \tag{6}$$

   where ∂X and ∂Y are the boundaries of regions X and Y respectively. The boundary pixels are defined as those with at least one neighbouring pixel outside the region [4].

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.6.5 Hausdorff Distance 95% Percentile

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.6.6 Average Symmetric Surface Distance

a. *Metric definition and/or formula:* With $d(a, B)$ defined by Equation 5 the Average Symmetric Surface Distance is defined as

$$\mathrm{ASSD}(X, Y) = \frac{\sum_{x \in \partial X} d(x, \partial Y) + \sum_{y \in \partial Y} d(y, \partial X)}{|\partial X| + |\partial Y|} \tag{7}$$

where ∂X and ∂Y are the boundaries of regions X and Y respectively. The boundary pixels are defined as those with at least one neighbouring pixel outside the region [4].

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.6.7 Mean Average Surface Distance

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.6.8 Boundary IoU

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.6.9 Absolute/Relative Volume Error

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.6.10 Symmetric Relative Volume Difference

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.6.11 Centerline Coefficient

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.6.12 Topology Precision

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.6.13 Topology Sensitivity

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.6.14 Surface Overlap

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.6.15 Surface Dice

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.6.16 Volumetric Dice

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.7 Instance segmentation

Basic task description.

2.7.1 Panoptic Quality

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.8 Image Regression

Basic task description.

2.8.1 Mean Absolute Error, Mean Squared Error, Root Mean Squared Error, Mean Absolute Percentage Error

a. *Metric definition and/or formula:*

   See formula in A.2.4 for Mean Absolute Error, in A.2.1 for Mean Squared Error, in A.2.2 for Root Mean Squared Error, in A.2.5 for Mean Absolute Percentage Error

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.8.2 R^2 Error

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.8.3 Root Mean Squared Log Error

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.8.4 Symmetric Mean Absolute Percentage Error

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.8.5 Mean Directional Accuracy

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.8.6 Median Absolute Error

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.8.7 Explained Variance Score

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.9 Panoptic segmentation

Basic task description.

2.9.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.10 Face analysis

Basic task description.

2.10.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.11 Saliency detection

Basic task description.

2.11.1 Increase in Confidence (IIC)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.11.2 Average Drop (AD)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.11.3 Average Drop in Deletion

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.11.4 Deletion Area Under Curve (DAUC)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.11.5 Insertion Area Under Curve (IAUC)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.11.6 Deletion Correlation (DC)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.11.7 Insertion Correlation (IC)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.12 Scene graph generation

Basic task description.

2.12.1 Recall@K

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.12.2 Mean Recall@K

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.12.3 No Graph Constraint Recall@K

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.13 Instance identification

Basic task description.

2.13.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.14 Feature extraction

Basic task description.

2.14.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.15 Geometric feature detection

Basic task description.

2.15.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.16 Image registration

Transforming multiple sets of image data into one common coordinate system.

2.16.1 Mean Squared Error

a. *Metric definition and/or formula:*

   See formula in A.2.1

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.16.2 Mutual Information

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.16.3 Normalized Mutual Information (NMI)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.16.4 Normalized cross correlation (NCC)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.16.5 Modality Independent Neighborhood Descriptor

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.17 Image stitching

Basic task description.

2.17.1 Peak Signal-to-Noise Ratio (PSNR)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.17.2 Structure Similarity Index Measurement (SSIM)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.18 Image similarity

Basic task description.

2.18.1 Mean Absolute Error, Mean Squared Error, Root Mean Squared Error, Normalized Mean Squared Error

a. *Metric definition and/or formula:*

See formula in A.2.4 for Mean Absolute Error, in A.2.1 for Mean Squared Error, in A.2.2 for Root Mean Squared Error, in A.2.3 for Normalized Mean Squared Error

In the context of image similarity, for two $m \times n$ images each with c channels, the Mean Absolute Error can be written as:

$$\mathrm{MAE} = \frac{1}{mnc} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} \sum_{k=0}^{c-1} \left| I_1(i,j,k) - I_2(i,j,k) \right| . \tag{8}$$

where indices i, j denote pixel coordinates and k denotes channel number.

Similarly the Mean Squared Error formula is

$$\mathrm{MSE} = \frac{1}{mnc} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} \sum_{k=0}^{c-1} \left[I_1(i,j,k) - I_2(i,j,k) \right]^2 \tag{9}$$

and the Root Mean Squared Error is

$$\mathrm{RMSE} = \sqrt{\frac{1}{mnc} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} \sum_{k=0}^{c-1} \left[I_1(i,j,k) - I_2(i,j,k) \right]^2} \tag{10}$$

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.18.2 Peak Signal-to-Noise Ratio (PSNR)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.18.3 Structure Similarity Index Measurement (SSIM)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.18.4 Multi-Scale Structural Similarity index (MS-SSIM)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.18.5 Complex Wavelet Structural Similarity index (CW-SSIM)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.18.6 Feature Similarity

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.18.7 Universal Image Quality Index

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.18.8 Learned Perceptual Image Patch Similarity (LPIPS)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.18.9 Deep Image Structure and Texture Similarity (DISTS)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.18.10 Information theoretic-based Statistic Similarity Measure (ISSM)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.18.11 Signal to reconstruction error ratio (SRE)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.18.12 Pearson Correlation Coefficient (PCC)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.18.13 Mutual Information

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.18.14 Normalized Mutual Information (NMI)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.18.15 Spectral Angle Mapper

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

2.19 Image retrieval (including cross modal retrieval)

Basic task description.

2.19.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

3 Spatial Analysis Tasks

3.1 Depth estimation

Basic task description.

3.1.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

3.2 Object / 3D reconstruction

Basic task description.

3.2.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

3.3 Object pose estimation

Basic task description.

3.3.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

3.4 Camera pose estimation

Basic task description.

3.4.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

3.5 Structure from Motion

Basic task description.

3.5.1 Absolute Trajectory Error

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

3.5.2 Relative Pose Error

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

3.6 SLAM

Simultaneously contructing or updating a map of an environment and estimating the robot's state (position, orientation, velocity) within said environment.

3.6.1 Metric

3.6.2 Absolute Tracjectory Error

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

3.6.3 Relative Pose Error

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

3.6.4 Loop Closure Precision and Recall

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

3.6.5 Map Consistency

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

3.6.6 Map Density and Coverage

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

4 Temporal analysis

4.1 Optical flow estimation

Basic task description.

4.1.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

4.2 Motion detection

Basic task description.

4.2.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

4.3 Single object tracking

Basic task description.

4.3.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

4.4 Multiple object tracking

Basic task description.

4.4.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

4.5 Action recognition

Basic task description.

4.5.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

4.6 Event detection

Basic task description.

4.6.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

4.7 (Multiple) Object re-identification

Basic task description.

4.7.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

4.8 Visual Odometry

Estimate the state (position, orientation, velocity) of a robot within an environment using camera measurements. Can be used as part of Visual Simultaneous Location and Mapping.

4.8.1 Absolute Tracjectory Error

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

4.8.2 Relative Pose Error

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

4.9 Change detection

Basic task description.

4.9.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

4.10 Deep fake detection

Basic task description.

4.10.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

4.11 Video summarisation

Basic task description.

4.11.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5 Image Generation Tasks

5.1 Image creation

Basic task description.

5.1.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.2 Image denoising

Basic task description.

5.2.1 Mean Squared Error, Root Mean Squared Error

a. *Metric definition and/or formula:*

   See formula in A.2.1 for Mean Squared Error, in A.2.2 for Root Mean Squared Error

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.2.2 Peak Signal-to-Noise Ratio (PSNR)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.2.3 Structure Similarity Index Measurement (SSIM)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.2.4 Multi-Scale Structural Similarity index (MS-SSIM)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.2.5 Feature Similarity

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.2.6 Universal Image Quality Index

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.2.7 Deep Image Structure and Texture Similarity (DISTS)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.2.8 Information theoretic-based Statistic Similarity Measure (ISSM)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.2.9 Signal to reconstruction error ratio (SRE)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.2.10 Visual information fidelity (VIF)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.2.11 Blind/Referenceless Image Spatial Quality Evaluator (BRISQUE)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.2.12 Natural Image Quality Evaluator (NIQE)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.2.13 Perception-based Image Quality Evaluator (PIQE)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.2.14 Model Specialization Metric (MSM)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.3 Super-resolution

Basic task description.

5.3.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.4 Image inpainting

Basic task description.

5.4.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.5 Image-to-image translation

Basic task description.

5.5.1 Mean Absolute Error, Mean Squared Error, Normalized Mean Squared Error

a. *Metric definition and/or formula:*

   See formula in A.2.4 for Mean Absolute Error, in A.2.1 for Mean Squared Error, in A.2.3 for Normalized Mean Squared Error

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.5.2 Peak Signal-to-Noise Ratio (PSNR)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.5.3 Structure Similarity Index Measurement (SSIM)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.5.4 Multi-Scale Structural Similarity index (MS-SSIM)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.5.5 Complex Wavelet Structural Similarity index (CW-SSIM)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.5.6 Learned Perceptual Image Patch Similarity (LPIPS)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.5.7 Deep Image Structure and Texture Similarity (DISTS)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.5.8 Pearson Correlation Coefficient (PCC)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.5.9 Normalized Mutual Information (NMI)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.5.10 FCNScore

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.5.11 SAMScore

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.5.12 Mean Blur (MB)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.5.13 Blur Ratio (BR)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.5.14 Natural Image Quality Evaluator (NIQE)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.5.15 Blur Effect (BE)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.5.16 Variance of Laplacian

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.5.17 Blurred Edge Widths (BEW)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.5.18 Cumulative Probability of Blur Detection (CPBD)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.5.19 Mean Line Correlation (MLC)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.5.20 Mean Shifted Line Correlation (MSLC)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.5.21 Just Noticeable Blur (JNB)

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.6 Style transfer

Basic task description.

5.6.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.7 Object swapping

Basic task description.

5.7.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

5.8 Novel view synthesis

Basic task description.

5.8.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

6 Video Generation Tasks

6.1 Video enhancement

Basic task description.

6.1.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

6.2 Video stabilisation

Basic task description.

6.2.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

6.3 Talking head generation

Basic task description.

6.3.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

7 Multi-modal Tasks

7.1 Visual question answering

Basic task description.

7.1.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

7.2 Image captioning

Basic task description.

7.2.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

7.3 Visual grounding

Basic task description.

7.3.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

7.4 RGB-depth fusion

Basic task description.

7.4.1 Metric

a. *Metric definition and/or formula:*

b. *Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:*

c. *Typical range of values obtained with this metric on state-of-the-art systems for this particular task:*

d. *Any other issues or subtleties that should be mentioned regarding this metric when applied to this task:*

A Base formulas

A.1 Formulas based on TP / FP / TN / FN

A.1.1 Accuracy

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{11}$$

A.1.2 Precision = Positive Predictive Value

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{12}$$

A.1.3 Recall = Sensitivity = True Positive Rate

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{13}$$

A.1.4 Specificity = Selectivity = True Negative Rate

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \tag{14}$$

A.1.5 False Positive Rate

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}} \tag{15}$$

A.1.6 False Negative Rate

$$\text{FNR} = \frac{\text{FN}}{\text{TP} + \text{FN}} = 1 - \mathit{Recall} \tag{16}$$

A.1.7 Negative Predictive Value

$$\text{NPV} = \frac{\text{TN}}{\text{TN} + \text{FN}} \tag{17}$$

A.1.8 F1 score

$$\begin{aligned} F_1 &= \frac{2}{\text{Recall}^{-1} + \text{Precision}^{-1}} && (18) \\ &= 2\,\frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} && (19) \\ &= \frac{2\ \text{TP}}{2\ \text{TP} + \text{FP} + \text{FN}} && (20) \end{aligned}$$

A.1.9 F_β score

$$\begin{aligned} F_\beta &= \frac{\beta^2 + 1}{\beta^2 \cdot \text{Recall}^{-1} + \text{Precision}^{-1}} && (21) \\ &= \frac{(1 + \beta^2) \cdot \text{Precision} \cdot \text{Recall}}{\beta^2 \cdot \text{Precision} + \text{Recall}} && (22) \\ &= \frac{(1 + \beta^2) \cdot \text{TP}}{(1 + \beta^2) \cdot \text{TP} + \text{FP} + \beta^2 \cdot \text{FN}} && (23) \end{aligned}$$

Any further details, hyperparameters, processing steps, underspecification of the formula, context dependencies, or other ambiguities in the metric itself that can affect reproducibility:

- Hyperparameter β which controls the relative importance of Recall versus Precision. $\beta = 1$ weights them equally and reduces F_β to F_1-score as above; $\beta > 1$ gives more importance to Recall; and $\beta < 1$ gives more importance to Precision.

A.1.10 Balanced accuracy

$$BalancedAccuracy = \frac{Sensitivity + Specificity}{2} \tag{24}$$

A.1.11 Matthews Correlation Coefficient

$$MCC = \frac{TN \cdot TP - FN \cdot FP}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{25}$$

A.1.12 Positive Likelihood Ratio

$$LR+ = \frac{Sensitivity}{1 - Specificity} \tag{26}$$

A.1.13 Negative Likelihood Ratio

$$LR- = \frac{1 - Sensitivity}{Specificity} \tag{27}$$

A.1.14 Youden's Index = Youden's J statistic = Bookmaker Informedness

$$J = Sensitivity + Specificity - 1 \tag{28}$$

A.2 Regression formulas

A.2.1 Mean Squared Error = Mean Squared Deviation

$$MSE = \frac{\sum_i (y_i - p_i)^2}{n} \tag{29}$$

where y_i and p_i are the true and predicted values for the ith data point and n is the number of data points

A.2.2 Root Mean Squared Error = Root Mean Squared Deviation

$$RMSE = \sqrt{\frac{\sum_i (y_i - p_i)^2}{n}} \tag{30}$$

where y_i and p_i are the true and predicted values for the ith data point and n is the number of data points

A.2.3 Normalized Mean Squared Error

$$NMSE = \frac{\sum_i (y_i - p_i)^2}{\sum_i (y_i - m)^2} \tag{31}$$

where y_i and p_i are the true and predicted values for the ith data point, n is the number of data points and $m = \frac{\sum y_i}{n}$

A.2.4 Mean Absolute Error

$$MAE = \frac{\sum_i |y_i - p_i|}{n} \tag{32}$$

where y_i and p_i are the true and predicted values for the ith data point and n is the number of data points

A.2.5 Mean Absolute Percentage Error

$$MAPE = \frac{1}{n}\sum_i \left|\frac{y_i - p_i}{y_i}\right| \cdot 100 \tag{33}$$

where y_i and p_i are the true and predicted values for the ith data point and n is the number of data points

A.2.6 R-Squared = R^2 = Coefficient of determination

$$R2 = 1 - \frac{\sum_i (y_i - p_i)^2}{\sum_i (y_i - m)^2} \tag{34}$$

where y_i and p_i are the true and predicted values for the ith data point, n is the number of data points and $m = \frac{sumyi}{n}$

A.2.7 Adjusted R-Squared

$$Adjusted\_R2 = 1 - \frac{\sum_i (y_i - p_i)^2/(n-1)}{\sum_i (y_i - m)^2/(n-k-1)} \tag{35}$$

where y_i and p_i are the true and predicted values for the ith data point, n is the number of data points, $m = \frac{\sum_i y_i}{n}$ and k is the number of independent variables

A.3 Intersection-over-Union, aka Jaccard Index

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} \tag{36}$$

$$= \frac{|A \cap B|}{|A| + |B| - |A \cap B|} \tag{37}$$

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